Hamza Idrees Backend Engineer

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Technical Skills

Languages: JavaScript, TypeScript, Python
Backend: Node.js, Express.js, NestJS, FastAPI, REST APIs, WebSockets, Swagger/OpenAPI, BullMQ
Databases: PostgreSQL, Sequelize, Prisma, PostGIS, MongoDB, Mongoose, Redis, Qdrant, Pinecone
Cloud and DevOps: AWS (EC2, RDS, ECR, S3, Lambda, SQS, CloudWatch), Docker, CI/CD, CodePipeline, CodeBuild, Git, Linux
Integrations: Firebase FCM, WhatsApp Cloud API, Chatwoot, Google APIs, Datashake, SendGrid, DeepL
Security & Testing: JWT, OAuth 2.0, RBAC, Rate Limiting, Input Validation, Webhook Verification, Jest, Mocha

Education

Bachelor of Science in Software Engineering, Islamabad, Pakistan
National University of Computer and Emerging Sciences (FAST-NUCES)
Leadership Roles & Activities

- **Daira Event Management Team** - Coordinated on-ground operations for an inter-university event covering games, debates, and student competitions.
- **Symphonic Society Member** - Helped organize student singing auditions and coordinated event operations for university music performances.

Work Experience

Global Software Consulting, *Backend Engineer* 12/2024 – Present
Lahore, Pakistan

- Developed and optimized multi-tenant REST APIs using PostgreSQL, MongoDB, and Redis, adding caching to high-traffic endpoints to reduce repeated database queries and load.
- Built an AWS Lambda ingestion pipeline that processes 40,000+ reviews within the 15-minute execution limit using optimized batching and pagination.
- Reduced AI content-generation time by 90%+ by building a batched OpenAI and Anthropic Claude pipeline that processes 100+ posts in under 30 seconds.
- Built and maintained 10+ BullMQ/Redis background queues for data ingestion, analysis, report generation, email delivery, and third-party platform synchronization.
- Implemented JWT authentication, RBAC, OAuth 2.0, rate limiting, input validation, and webhook signature verification, while maintaining 90%+ automated test coverage with Jest, Mocha, and Supertest.
- Containerized and deployed a NestJS/Turborepo backend on AWS EC2 using Docker, RDS PostgreSQL/PostGIS, Redis, and S3, with automated deployments through CodePipeline, CodeBuild, ECR, SSM, and Prisma migrations.
- Built a real-time emergency-alert flow using BullMQ, Socket.IO, and Firebase Cloud Messaging to deliver authenticated location updates and push notifications.
- Built a production AI receptionist for Rosa (Solar Chiropractic) that automates appointment booking, rescheduling, cancellation, and operational follow-ups across WhatsApp and email through PracticeHub.
- Automated The Nanny Line's end-to-end booking workflow across WhatsApp and email, covering request capture, pricing, nanny matching, availability, reconfirmation, and Google Calendar scheduling.

Projects

Obenan - Multi-Tenant SaaS Platform ,

Node.js, Express.js, PostgreSQL, Sequelize, MongoDB, Redis, BullMQ, AWS, JWT, OAuth 2.0

- Multi-tenant SaaS platform for business-location management and online reputation, supporting 400+ businesses across 1,000+ locations in 35+ countries with integrations across Google, Facebook, and external review platforms.
- Designed and maintained a dual-database architecture using PostgreSQL for relational business data and MongoDB for document-based workloads, with 100+ Sequelize models and Redis caching for frequently accessed data.
- Built backend workflows for review management, reporting, location synchronization, and AI-assisted content generation, with role-based access, OAuth 2.0, validation, rate limiting, and webhook verification.

Reddr - Personal Safety and Real-Time Alert Platform ,

NestJS, PostgreSQL, Prisma, PostGIS, Redis, BullMQ, Socket.IO, Docker, AWS

- Personal-safety platform that connects mobile panic events with backend emergency workflows, live location tracking, notifications, and a real-time map.
- Designed the backend around NestJS and Prisma with PostgreSQL/PostGIS for location-aware data, Redis/BullMQ for background processing, and Socket.IO for real-time communication.
- Implemented authenticated REST and WebSocket interfaces used by the mobile application to update live location state, trigger emergency workflows, and receive push notifications.

The Nanny Line - AI Booking Automation Platform,

N8N, OpenAI, Anthropic Claude, JavaScript, Google Sheets/Calendar APIs, WhatsApp Cloud API, Chatwoot, Gmail

- AI-powered booking system that handles nanny requests across WhatsApp and email and manages the workflow from initial request through confirmation and scheduling.
- Designed conversation-level booking state in Google Sheets with deduplication and structured tracking across multiple communication channels.
- Built a deterministic pricing engine covering location-based rates, holiday pricing, overnight and one-way travel compensation, VAT, and agency commission.
- Implemented location/language matching, nanny availability broadcasts, response collection, nanny selection, 24-hour reconfirmation, Google Calendar scheduling, and safeguards for API limits and workflow loops.

Rosa (Solar Chiropractic) - AI Receptionist and Practice Automation,

N8N, OpenAI, Anthropic Claude, JavaScript, PracticeHub API, Pinecone, WhatsApp Cloud API, Chatwoot, Gmail

- AI receptionist and practice-automation system connected to PracticeHub, WhatsApp, and email for appointment-related conversations and operational workflows.
- Designed deterministic booking rules for same-day restrictions, recurring reviews, reactivation cases, and family/on-behalf bookings, with a Pinecone-based RAG layer and contact-level privacy controls.
- Automated confirmations, reminders, recalls, payment and no-show workflows, scheduled practice reports, retry handling, and escalation to staff when human input is required.

Lumen - Clinical RAG Assistant

Python, FastAPI, Qdrant, OpenAI, SSE, Next.js 14, Tailwind CSS

- Built a clinical search and question-answering system that lets users query patient data in natural language and receive responses grounded in retrieved records.
- Designed an end-to-end RAG pipeline using GPT-4o-mini for intent extraction, OpenAI embeddings, Qdrant vector search, and SSE streaming, achieving sub-100ms vector-search latency and under-2-second first-token response time.
- Reduced irrelevant retrievals by 60%+ with a 20+ filter query parser covering condition, age, doctor, dates, billing, and blood type; added similarity-threshold checks and fallback handling for low-confidence queries while supporting bulk ingestion of 5,000 records in under 60 seconds.